$$u(x_1, x_2) = K x_1^{\alpha} x_2^{1-\alpha}$$

s.t. $\quad P_1 x_1 + P_2 x_2 = w$

Setting up lagrangian:

$$L(x_1, x_2; \lambda) = K x_1^{\alpha} x_2^{1-\alpha} + \lambda(w - P_1 x_1 - P_2 x_2)$$

FOCs:

$$\frac{\delta L}{\delta x_1} = K \alpha x_1^{\alpha-1} x_2^{1-\alpha} - \lambda P_1 = 0 \quad \text{---} \quad (i)$$

$$\frac{\delta L}{\delta x_2} = K x_1^{\alpha} (1-\alpha) x_2^{-\alpha} - \lambda P_2 = 0 \quad \text{---} \quad (ii)$$

$$\frac{\delta L}{\delta \lambda} = w - P_1 x_1 - P_2 x_2 = 0 \quad (iii)$$

From (i)

$$\lambda P_1 = K \alpha x_1^{\alpha-1} x_2^{1-\alpha} \quad \text{----} \quad (iv)$$

from (ii)

$$\lambda P_2 = K(1-\alpha) x_1^{\alpha} x_2^{-\alpha} \quad \text{---} \quad (v)$$

(iii) ÷ (iv)

$$\frac{P_1}{P_2} = \frac{\alpha}{(1-\alpha)} \cdot \frac{x_2}{x_1} \qquad (v)$$

From (v)

$$P_1 x_1 = \frac{\alpha}{(1-\alpha)} P_2 x_2 \qquad --- \quad (vi)$$

and, $$P_2 x_2 = \frac{(1-\alpha)}{\alpha} P_1 x_1 \qquad --- \quad (vii)$$

Substituting (vii) into budget constraint:

$$P_1 x_1 + \frac{(1-\alpha)}{\alpha} P_1 x_1 = w$$

$$\Rightarrow P_1 x_1 \left(1 + \frac{1-\alpha}{\alpha}\right) = w$$

$$\Rightarrow \frac{1}{\alpha} P_1 x_1 = w$$

$$\therefore x_1 = \frac{\alpha w}{P_1}$$

Substituting (vi) into budget constraint (iii)

$$\frac{\alpha}{(1-\alpha)} P_2 x_2 + P_2 x_2 = w$$

$$\Rightarrow \left(\frac{\alpha}{1-\alpha} + 1\right) P_2 x_2 = w$$

$$\Rightarrow \frac{1}{(1-\alpha)} \cdot P_2 x_2 = w$$

$$\Rightarrow x_2 = \frac{w(1-\alpha)}{P_2}$$

$\therefore$ walrasian demand for good 1, $x_1 (P_1, w) = \frac{\alpha w}{P_1}$

$\therefore$ walrasian demand for good 2, $x_2 (P_2, w) = \frac{(1-\alpha) w}{P_2}$

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ii) Indirect utility function:

$$v(P, \omega) = K\left(\frac{\alpha \omega}{P_1}\right)^{\alpha}\left(\frac{(1-\alpha)\omega}{P_2}\right)^{1-\alpha}$$

$$= K\omega\left(\frac{\alpha}{P_1}\right)^{\alpha}\left(\frac{(1-\alpha)}{P_2}\right)^{1-\alpha}$$

According to Roy's identity:

$$-\frac{\frac{\delta v(P, \omega)}{\delta P_k}}{\frac{\delta v(P, \omega)}{\delta \omega}} = x_k(P, \omega) \quad \text{for every good } k.$$

Now, for good 1:

$$\frac{\delta v(P, \omega)}{\delta P_1} = \frac{\delta}{\delta P_1}\left(K\omega\alpha^{\alpha} P_1^{-\alpha}(1-\alpha)^{1-\alpha} P_2^{\alpha-1}\right)$$

$$= -\alpha K\omega\alpha^{\alpha} P_1^{-\alpha-1}(1-\alpha)^{1-\alpha} P_2^{\alpha-1}$$

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$$\frac{\delta v(p,w)}{\delta w} = \frac{\delta}{\delta w}\left(kw\alpha^{\alpha} p_1^{-\alpha}(1-\alpha)^{1-\alpha} p_2^{\alpha-1}\right)$$

$$= k\alpha^{\alpha} p_1^{-\alpha}(1-\alpha)^{1-\alpha} p_2^{\alpha-1}$$

$$\therefore \frac{\frac{\delta v(p,w)}{\delta p_1}}{\frac{\delta v(p,w)}{\delta w}} = \frac{-\alpha k w \alpha^{\alpha} p_1^{-\alpha-1}(1-\alpha)^{1-\alpha} p_2^{\alpha-1}}{k\alpha^{\alpha} p_1^{-\alpha}(1-\alpha)^{1-\alpha} p_2^{\alpha-1}}$$

$$-\frac{\frac{\delta v(p,w)}{\delta p_1}}{\frac{\delta v(p,w)}{\delta w}} = \frac{\alpha w p_1^{-\alpha-1}}{p_1^{-\alpha}}$$

$$= \frac{\alpha w}{p_1}$$

$$= x_1(p_1, w)$$

Roy's identity satisfied.

Now for good 2:

$$\frac{\delta v(P,w)}{\delta P_2} = \frac{\delta}{\delta P_2}\left(k w \alpha^{\alpha} P_1^{-\alpha}(1-\alpha)^{1-\alpha} P_2^{\alpha-1}\right)$$

$$= (\alpha-1) k w \alpha^{\alpha} P_1^{-\alpha}(1-\alpha)^{(1-\alpha)} P_2^{(\alpha-2)}$$

$$\frac{\frac{\delta v(P,w)}{\delta P_2}}{\frac{\delta v(P,w)}{\delta w}} = \frac{(\alpha-1) k w \alpha^{\alpha} P_1^{-\alpha}(1-\alpha)^{(1-\alpha)} P_2^{(\alpha-2)}}{k \alpha^{\alpha} P_1^{-\alpha}(1-\alpha)^{1-\alpha} P_2^{(\alpha-1)}}$$

$$-\frac{\frac{\delta v(P,w)}{\delta P_2}}{\frac{\delta v(P,w)}{\delta w}} = \frac{-(\alpha-1) w P_2^{\alpha-2}}{P_2^{\alpha-1}}$$

$$= (1-\alpha) w P_2^{-1}$$

$$= \frac{(1-\alpha) w}{P_2} = x_2(P_2, w)$$

$\therefore$ Roy's identity satisfied.

iii) min $P_1x_1 + P_2x_2$

s.t. $kx_1^{\alpha}x_2^{1-\alpha} \geq u$

Setting up the Lagrangian:

$$L = P_1x_1 + P_2x_2 + \mu\left(u - kx_1^{\alpha}x_2^{1-\alpha}\right)$$

FOCs:

$$\frac{\delta L}{\delta x_1} = P_1 - \mu \alpha k x_1^{\alpha-1} x_2^{1-\alpha} \geq 0$$

$$\frac{\delta L}{\delta x_2} = P_2 - \mu(1-\alpha) k x_1^{\alpha} x_2^{-\alpha} \geq 0$$

$$\frac{\delta L}{\delta \mu} = u - kx_1^{\alpha}x_2^{1-\alpha} \geq 0$$

For interior solution, the FOCs become:

$$P_1 = u \alpha k x_1^{\alpha-1} x_2^{1-\alpha} \quad \text{--- (i)}$$

$$P_2 = u(1-\alpha) k x_1^{\alpha} x_2^{-\alpha} \quad \text{--- (ii)}$$

$$u = k x_1^{\alpha} x_2^{1-\alpha} \quad \text{--- (iii)}$$

(i) ÷ (ii)

$$\frac{P_1}{P_2} = \frac{u \alpha k x_1^{\alpha-1} x_2^{1-\alpha}}{u(1-\alpha) k x_1^{\alpha} x_2^{-\alpha}}$$

$$\frac{P_1}{P_2} = \frac{\alpha x_1^{\alpha-1} x_2^{1-\alpha}}{(1-\alpha) x_1^{\alpha} x_2^{-\alpha}}$$

$$\frac{P_1}{P_2} = \frac{\alpha}{(1-\alpha)} \frac{x_2}{x_1} \quad \text{--- (iv)}$$

From (iv)

$$x_1 = \frac{\alpha P_2 x_2}{(1-\alpha) P_1} \quad \text{--- (v)}$$

and, $$x_2 = \frac{(1-\alpha) P_1 x_1}{\alpha P_2} \quad \text{--- (vi)}$$

Substituting (vi) into (iii)

$$K x_1^{\alpha} \left(\frac{(1-\alpha) P_1 x_1}{\alpha P_2} \right)^{1-\alpha} = u$$

$$x_1^{\alpha} \left(\frac{(1-\alpha) P_1}{\alpha P_2} \right)^{1-\alpha} x_1^{1-\alpha} = \frac{u}{K}$$

$$\Rightarrow x_1^{\alpha+1-\alpha} = \frac{u}{K} \left(\frac{\alpha P_2}{(1-\alpha) P_1} \right)^{1-\alpha}$$

$$\therefore x_1^{*} = \frac{u}{K} \left(\frac{\alpha}{(1-\alpha)} \frac{P_2}{P_1} \right)^{1-\alpha}$$

$$\therefore h_1(P,\omega) = \frac{u}{K}\left(\frac{\alpha}{(1-\alpha)}\frac{P_2}{P_1}\right)^{1-\alpha}$$

Substituting (v) into (iii)

$$K\left(\frac{\alpha P_2 x_2}{(1-\alpha) P_1}\right)^{\alpha} x_2^{1-\alpha} = u$$

$$x_2^{1-\alpha} \cdot x_2^{\alpha} = \frac{u}{K}\left(\frac{(1-\alpha)}{\alpha}\frac{P_1}{P_2}\right)^{\alpha}$$

$$\therefore x_2^{*} = \frac{u}{K}\left(\frac{(1-\alpha)}{\alpha}\frac{P_1}{P_2}\right)^{\alpha}$$

$$x_2^{*} = h_2(P,\omega) = \frac{u}{K}\left(\frac{(1-\alpha)}{\alpha}\frac{P_1}{P_2}\right)^{\alpha}$$